Question ID: b64e2c7f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

Monarch butterflies can fly only with a body temperature of at least $55.0 \text{ degrees Fahrenheit } (°\text{F})$. If a monarch butterfly's body temperature is $51.3°\text{F}$, what is the minimum increase needed in its body temperature, in $°\text{F}$, so that it can fly?

Answer

A. 1.3

B. 3.7

C. 5.0

D. 6.3

Correct Answer: B

Rationale

Choice B is correct. It's given that monarch butterflies can fly only with a body temperature of at least $55.0 \text{ degrees Fahrenheit } (°\text{F})$. Let x represent the minimum increase needed in the monarch butterfly's body temperature to fly. If the monarch butterfly's body temperature is $51.3 °\text{F}$, the inequality $51.3 + x \geq 55.0$ represents this situation. Subtracting 51.3 from both sides of this inequality yields $x \geq 3.7$. Therefore, if the monarch butterfly's body temperature is $51.3°\text{F}$, the minimum increase needed in its body temperature, in $°\text{F}$, so that it can fly is 3.7.

Choice A is incorrect. This is the minimum increase needed in body temperature if the monarch butterfly's body temperature is $53.7°\text{F}$, not $51.3 °\text{F}$.

Choice C is incorrect. This is the minimum increase needed in body temperature if the monarch butterfly's body temperature is $50.0°\text{F}$, not $51.3 °\text{F}$.

Choice D is incorrect. This is the minimum increase needed in body temperature if the monarch butterfly's body temperature is $48.7°\text{F}$, not $51.3 °\text{F}$.